Caption A:

CellForge home-turf transfer checks. CellScientist is evaluated on four CellForge/scPerturb tasks (Norman, Schiebinger, Papalexi RNA, Papalexi Protein) under the same six-metric family, showing strong transfer evidence beyond the shared BBBC021 setting.

Method	Dataset	MSE ↓	PCC ↑	R^2 ↑	MSE_DE ↓	PCC_DE ↑	R^2_DE ↑
Gene Knock Out Perturbation – scRNAseq Dataset							
Norman	CellForge	0.0034±0.0023	**0.9846±0.0418**	0.9609±0.0081	**0.1736±0.0677**	0.8109±0.0133	0.5975±0.0539
	CellScientist	**0.0033±0.0019**	0.9871±0.03600	**0.9675±0.0035**	0.1846±0.0293	**0.9060±0.0512**	**0.6328±0.0479**
Cytokine Perturbation – scRNA-seq Dataset							
Schiebinger	CellForge	0.0428±0.0205	0.5697±0.0943	0.5043±0.0541	**0.0144±0.0349**	0.3396±0.0403	0.2832±0.1154
	CellScientist	**0.0427±0.0512**	**0.8818±0.0181**	**0.6026±0.0094**	0.0481±0.0151	**0.8935±0.0118**	**0.5793±0.0134**
Gene Knock Out Perturbation – scCITEseq (RNA) Dataset							
Papalexi RNA	CellForge	0.0417±0.0051	0.6935±0.1995	0.3687±0.0651	0.0535±0.1566	0.6406±0.1940	0.2354±0.0224
	CellScientist	**0.0283±0.0058**	**0.8193±0.0035**	**0.5134±0.0176**	**0.0231±0.0056**	**0.8311±0.0029**	**0.5583±0.0203**
Gene Knock Out Perturbation – scCITEseq (Protein) Dataset							
Papalexi Protein	CellForge	0.0070±0.0387	0.7495±0.0653	**0.6872±0.0956**	0.2921±0.0045	0.7409±0.0970	0.5489±0.0749
	CellScientist	**0.00651±0.0264**	**0.7941±0.0358**	0.6758±0.0184	**0.1914±0.0026**	**0.8275±0.0108**	**0.7441±0.0030**

Caption B:

Matched-budget fairness pilot on BBBC021. Strong baselines are tuned under a shared 6-hour protocol to test whether budget-matched search qualitatively reverses the shared-setting conclusion.

Method	Wall-clock (h)	MSE↓	PCC↑	R^2↑	RMSE_DEG-20↓	RMSE_DEG-50↓	PCC_DEG-20↑	PCC_DEG-50↑
FLAML	4.45	0.1457±0.07	0.9631±0.02	**0.1101±0.17**	0.7002±0.18	0.5094±0.12	0.9715±0.01	0.9672±0.01
Random Search	6.27	0.3421±0.13	0.9091±0.03	0.1499±0.10	1.2338±0.31	0.8335±0.18	0.9149±0.04	0.9134±0.03
CellScientist	—	**0.1057±0.03**	**0.9735±0.01**	0.1053±0.09	**0.5844±0.11**	**0.4325±0.08**	**0.9818±0.01**	**0.9781±0.01**

Caption C:

System-level trade-off analysis reorganized from Table 5. Transition-wise deltas show how HRT, LCA, and PDR improve feasibility, compliance, and peak quality relative to naive retrying.

Transition	BBBC036 (ΔSR / ΔR-SR / ΔAvg PCC / ΔBest PCC / ΔTime)	BBBC047 (ΔSR / ΔR-SR / ΔAvg PCC / ΔBest PCC / ΔTime)
M0 → M1+ Naive re-prompting	-0.263 / -0.263 / +0.0799 / +0.0803 / +7.71h	-0.158 / -0.158 / +0.0845 / +0.0829 / +11.71h
M1 → M2+ HRT	**+0.684 / +0.421 / +0.0615 / +0.0565 / -7.62h**	**+0.731 / +0.378 / +0.0205 / +0.0194 / -11.33h**
M2 → M3+ LCA	**+0.053 / +0.053 / +0.0412 / +0.0164 / -1.88h**	**+0.059 / +0.096 / +0.0843 / +0.1187 / -2.58h**
M3 → M4+ PDR **(CellScientist)**	**0.000 / +0.052 / +0.0230 / +0.0474 / -0.53h**	**0.000 / +0.030 / +0.0224 / +0.0253 / -0.80h**

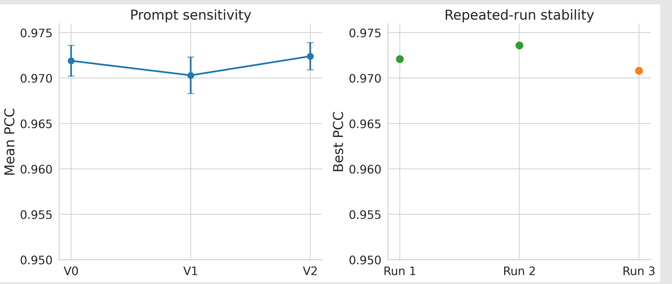
Caption D:

Cross-task system-level ablation on LINCS2020. The same M0–M4 closed-loop pattern is tested beyond morphology using success/compliance and average/best PCC metrics.

Variant	HRT	LCA	PDR	SR↑	R-SR↑	Best PCC↑
M0	✗	✗	✗	0.210	0.526	0.5934
M1	✗	✗	✗	0.526	0.588	0.6043
M2	✓	✗	✗	0.941	0.684	0.6123
M3	✓	✓	✗	0.947	0.714	0.6164
M4	✓	✓	✓	**1.000**	**0.789**	**0.6368**

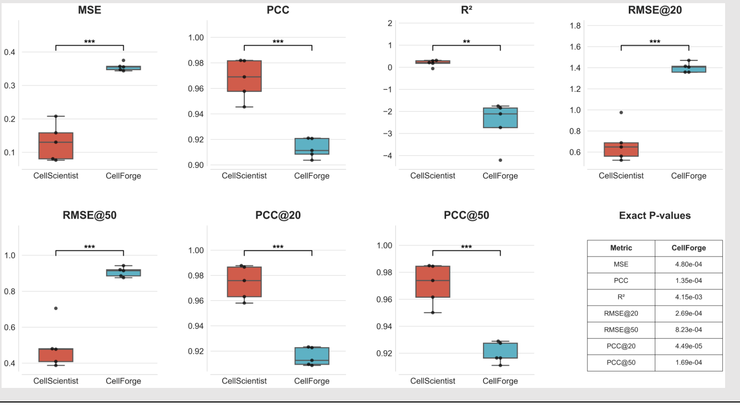
Caption E:

Prompt robustness and repeated-run stability on BBBC021. Small prompt paraphrases preserve the shared-setting conclusion, and repeated runs show stable behavior with explicit rollback of no-improvement updates.



Caption F:

Exact significance on the shared BBBC021 setting. Exact p-values are reported for the flagship CellScientist vs. CellForge comparison to statistically ground the main shared-setting result.



Metric	CellForge
MSE	4.80e-04
PCC	1.35e-04
R^2	4.15e-03
RMSE@20	2.69e-04
RMSE@50	8.23e-04
PCC@20	4.49e-05
PCC@50	1.69e-04